

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 4 Winter 2025 (Supply.)

Course : Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTMC401 - MANUFACTURING PROCESSES - I

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

1. Shrinkage in casting occurs due to: A) Expansion during solidification B) Density decrease C) Volume reduction during cooling D) Mold erosion
2. Fluidity of molten metal refers to: A) Its density B) Its viscosity C) Ability to flow and fill mold cavities D) Cooling rate
3. Which defect occurs due to non-uniform deformation in rolling? A) Lamination B) Edge cracks C) Blow holes D) Shrinkage cavity
4. Wire drawing is mainly used to reduce: A) Length B) Diameter C) Hardness D) Temperature
5. SMAW stands for: A) Submerged Metal Arc Welding B) Shielded Metal Arc Welding C) Solid Metal Arc Welding D) Special Metal Arc Welding
6. Spot welding is a type of: A) Arc welding B) Resistance welding C) Gas welding D) Brazing
7. Boring operation is used to: A) Create a new hole B) Enlarge an existing hole C) Cut threads D) Finish surfaces
8. Tapping operation is used for: A) Making slots B) Internal threading C) External threading D) Surface finishing
9. Milling uses: A) Single-point tool B) Multi-point cutting tool C) Abrasive wheel D) Laser
10. Broaching is a: A) Slow process B) High production process C) Casting process D) Joining process
11. Centrifugal casting is best suited for: A) Thin plates B) Hollow cylindrical parts C) Complex solid parts D) Small intricate components
12. Impression-die forging produces: A) Rough shapes B) Near-net shapes C) Hollow shapes only D) Sheets

Q2. Solve the following.

- A) Describe the process of solidification of pure metal and alloy with the help of appropriate diagrams. 6
- B) Explain open-die and closed-die forging operations using schematic diagrams. 6

- Q3. Solve the following.
- A) Explain boring, reaming, and tapping operations with the help of schematic diagrams. 6
 - B) Discuss different types of casting defects w.r.t. causes, remedies and schematic diagrams. 6
- Q4. Solve Any Two of the following.
- A) Explain investment casting process w.r.t. principle of working, steps involved, schematic diagrams, advantages, and applications. 6
 - B) Discuss oxy-fuel-gas welding with the help of schematic diagram. What are its advantages and limitations? 6
 - C) Explain lathe machine with neat sketch and give the description of its components. 6
- Q5. Solve Any Two of the following.
- A) Explain the principle of working of resistance welding using schematic diagram. Discuss spot welding and seam welding processes. 6
 - B) What are the various lathe operations? Explain each using schematic diagram. 6
 - C) Describe gear generating processes with the help of schematic diagrams. 6
- Q6. Solve Any Two of the following.
- A) What is extrusion? Explain hot extrusion, direct extrusion, and indirect extrusion processes using schematic diagrams. 6
 - B) Differentiate between hot rolling and cold rolling processes. Draw schematic diagram of flat rolling process. Explain the types of rolling mills using appropriate diagrams. 6
 - C) Discuss wire drawing process w.r.t. principle of working, schematic diagram, equipment details, advantages, limitations, and applications. 6

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